

SZABIST
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Final Year Project Long Proposal

Multi-Agent Market Research Platform

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Revision History

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1. Project Description

Professionals in outdoor advertising research pull data from scattered sources. Analysts process digital campaign logs, physical photographs, GPS coordinates, and paper surveys every day. Managing these disconnected pieces across multiple cities creates a massive administrative workload. Field staff currently run different software applications for different tasks. This disjointed approach makes it difficult to match a specific photograph to an exact street location. Office workers waste hours sorting loose files before they even start evaluating market trends. The industry requires a single platform to secure this evidence and drive the workflow directly from street capture to the final review.

The proposed system will operate as a central workspace to manage research from initial planning to the final report. The architecture will avoid bulky, all-in-one software designs. It will instead deploy a network of specialized AI agents to handle the workload. Specific agents will take over distinct tasks, like collecting field metrics, running quality checks, and drafting summaries. Each tool will finish its step and pass a clean data file to the next agent in the sequence. A central database will anchor these moving parts to track every source accurately. Whenever the automated tools spot messy data, the system will flag those items for manual human review. The platform will establish a reliable, end-to-end process built specifically for market analysis.

The new platform directly connects every research question to the raw field evidence and the final reports. This traceable workflow provides teams with a dependable method for market research. The unified architecture actively prevents common mistakes caused by manual data management. Companies experience real operational improvements from this direct connection. Field teams significantly reduce their administrative workloads. Managers verify the submitted data much faster. Analysts publish the final reports earlier.

The new system resolves immediate logistical bottlenecks. The platform also provides a flexible base for future projects. Organizations easily plug in new data streams without breaking the core pipeline. System administrators quickly assign different AI agents. This structural design allows agencies to scale operations safely. Research teams maintain complete visibility over their information. Executives base their final decisions on hard facts.

2. Real World Problem

The out-of-home (OOH) advertising industry generates massive amounts of data as it grows. This rapid expansion creates a strict demand for reliable evidence management. The sector reports a record $9.46 billion in revenue for 2025. This figure marks nineteen consecutive quarters of financial growth. Digital screens now account for 36.3% of that

total market. Brands spend heavily on these physical spaces. Advertising agencies require consistent and transparent campaign measurement across all media types to justify these brand investments. Geopath and the Out of Home Advertising Association of America currently manage a new audience measurement initiative launched in March 2026 to address these exact demands. The modern market simply needs stronger infrastructure to process complex performance data and hold operators accountable.

Within the target organization, these challenges are reflected in a research workflow in which survey responses, campaign details, field photographs, location information, spreadsheet calculations, and web references are maintained through separate activities. Researchers must manually reconcile these sources before comparing locations, time periods, or campaigns, while missing collection dates, duplicate responses, inconsistent naming, and unsupported information can complicate verification and analysis. The current process also requires separate effort for planning research, collecting information, extracting relevant content, checking data quality, verifying evidence, performing analysis, and preparing reports. These activities correspond to distinct functions that can be handled through specialized agents: a research planning agent can structure the research requirements; data collection agents can gather information from permitted sources; an information extraction agent can organize collected content; quality and verification agents can identify inconsistencies, duplicates, missing information, and unsupported claims; and analysis and reporting agents can process validated evidence and produce traceable findings.

If these gaps remain, the organization will face ongoing costs for manual data preparation, reconciliation, verification, and reporting. The likelihood of inconsistent or poorly supported findings will also increase as campaign volumes grow. Broader data-quality research will highlight the same concern: an IBM Institute for Business Value report published in 2026 will show that 43% of chief operating officers identify data quality as their main data priority, while more than one-quarter of organizations report annual losses above US$5 million due to poor data quality. For the target organization, the workload will provide a direct basis for estimating operational costs. A campaign covering 100 locations will require 50 staff-hours for manual reconciliation alone, based on 30 minutes per location, before verification and reporting are included. Higher campaign volumes will increase both labor demands and processing time.

3. Project Stream

✓	Web-Based FYP		Desktop Application
✓	Mobile App		Game
	Hardware-Based		

4. Modules

Table 1 provides a breakdown of the modules within the proposed system and the students assigned to each module based on the deadlines, clearly presenting the distribution of responsibilities across the development phases.

Table 1: Module Breakdown and Assignment

Deadline	**Modules**	**Student**
P1 Mid	User Management	Aroon Kumar (2312466)
	Questionnaire Management	Mudasir Ali (2312465)
	Campaign Management	Abdullah Zafar Khan (2312467)
P1 Final	Attendance Verification	Aroon Kumar (2312466)
	Data Synchronization	Mudasir Ali (2312465)
	Field Data-Collection	Abdullah Zafar Khan (2312467)
P2 Mid	AI Verification	Aroon Kumar (2312466)
	Market Intelligence	Mudasir Ali (2312465)
	Media Processing	Abdullah Zafar Khan (2312467)
P2 Final	Agent Orchestration	Aroon Kumar (2312466)
	Field Performance	Mudasir Ali (2312465)
	AI-Agent Analytics	Abdullah Zafar Khan (2312467)

5. Names of existing similar FYPs (if repeated)

None.

6. New Functionalities (main) to be added Features

N/A.

7. Development Environment

Tools:	Python, FastAPI, React, TypeScript, Flutter, Dart, LangGraph, Git, REST API client, Docker, browser developer tools
DBMS:	PostgreSQL, SQLite
Platform:	Visual Studio Code, Android Studio

8. Introduction

Market researchers collect and analyse information to answer business and market questions. Researchers collect primary data directly for a specific study through questionnaire responses, field observations, photographs, GPS records, and interviews. Researchers obtain secondary data from existing sources, including websites, reports, public databases, market publications, and competitor records. In outdoor advertising, researchers use both types of data to evaluate campaigns. Field workers record a billboard's location and condition during site visits, while questionnaires capture audience responses and secondary sources provide market or competitor information. The system organizes these different sources, verifies the collected information, and links each record to its original source.

The proposed platform provides a coordinated workflow for market research across multiple markets, campaigns, and locations. It manages campaigns and locations, research questionnaires, and Android-based field collection. The system also handles geofenced attendance and visit verification, offline data storage, incremental synchronization, media management, secondary data collection, quality checks, and cross-source verification. Specialized AI agents support research planning, data collection, information extraction, verification, analysis, dashboard preparation, reporting, and coding tasks. Selected AI models also run on mobile devices to perform preliminary analysis of images, audio, and field data. An analytics dashboard visualizes and compares collected data. Human reviewers remain involved in disputed or exceptional cases.

The text below previews the core industry challenges before setting up the boundaries and main goals of the proposed system. Subsequent chapters will break down the overall architecture, paying close attention to data syncing routines and AI agent pipelines. The documentation will then list the required development tools, quality checks, performance metrics, and the rollout timeline. To wrap up, a practical walkthrough will illustrate how the platform handles raw field evidence through automated pipelines to produce the final analytical reports.

9. Application Review

This section evaluates three established platforms: Qualtrics, SurveyMonkey, and ODK, to measure their standard capabilities. The review examines how each software handles routine questionnaires, field data collection without internet access, location tracking, and media uploads. Placing these existing features side-by-side with the requirements of the proposed system exposes clear technical limitations. Uncovering these missing functions clarifies the specific features and custom tools that the new architecture must build.

Qualtrics: Qualtrics provides web and mobile interfaces for designing, launching, and reviewing surveys. Field researchers use the Qualtrics Offline Surveys application on iOS and Android devices to download forms, capture entries offline, and upload them once connectivity returns. The software logs GPS coordinates with each submission and supplies basic visualization tools for initial summaries. However, official product documentation indicates that the platform misses core requirements for outdoor advertising workflows. The software does not provide registries for billboards and campaigns, geofenced attendance verification, or hierarchical site management. It also omits automated evidence validation, on-device artificial intelligence processing, pitch analysis, and coordinated research agents [1].

The main features of the Application are listed below:

- Web-based survey creation and distribution.
- Offline survey collection through mobile devices.
- Android and iOS support for offline collection.
- Geolocation data associated with survey responses.
- Device identification for collected responses.
- Survey response management and analysis.
- Results dashboards with charts, tables, filters, and visualizations.
- User permissions and controlled access to survey results.

The limitations of the application are listed below:

- The offline application does not support every survey feature.
- Access to offline collection depends on the required account permissions.
- The documented workflow focuses on survey collection and analysis rather than a dedicated billboard or campaign registry.
- The reviewed documentation does not describe geofenced attendance linked to assigned campaign locations.

- The reviewed documentation does not describe a dedicated workflow for campaign-linked photographs, audio, attendance evidence, and cross-source verification.
- The reviewed documentation does not establish mobile AI processing, billboard scoring, pitch analysis, or coordinated research and coding agents.
- Although dashboards and access controls are available, the reviewed documentation does not establish a field-agent performance dashboard with organization-defined scoring and reward workflows.

SurveyMonkey: SurveyMonkey manages questionnaire design, distribution, and response filtering, using web and mobile applications to display chart-based analytics. Teams perform offline field collection on iOS and Android devices through the Survey-Monkey anywhere mobile application. Users cache surveys before entering disconnected areas and sync data when the network signal returns. The platform handles basic field polling effectively, but subscription tiers restrict several advanced offline functions. The system also omits the specialized capabilities required for outdoor advertising research. The platform does not offer a dedicated billboard registry, geofenced visit verification, hierarchical location structuring, SQLite-based offline-first persistence with incremental sync, mobile AI processing, pitch analysis, or multi-agent research and coding workflows [2].

The main features of the Application are listed below:

- Questionnaire creation and distribution.
- Mobile survey administration.
- Offline response collection through SurveyMonkey Anywhere.
- Upload of collected responses after reconnection.
- Response status and collection metadata.
- Result filtering and comparison.
- Charts and survey-result visualization.
- Device management and user access controls.

The limitations of the application are listed below:

- Offline mode is restricted to compatible survey configurations and is not available for all survey features.
- Offline functionality requires the appropriate account access and configuration.
- The reviewed documentation does not describe a dedicated billboard or campaign registry.

- The reviewed documentation does not describe hierarchical market locations linked to assigned field visits.
- Standard offline survey collection does not establish the proposed geofenced attendance and visit-evidence workflow.
- The reviewed documentation does not establish the proposed SQLite-based offline-first architecture with incremental synchronization and explicit synchronization queues.
- The reviewed documentation does not establish mobile-side AI for image-quality analysis, billboard scoring, or audio processing.
- The reviewed documentation does not establish automated field-performance scoring, pitch analysis, or AI-agent orchestration.

ODK: Open Data Kit (ODK) facilitates structured field-data collection by pairing the ODK Collect Android app with a central management server, ODK Central. Field teams use ODK Collect to complete offline forms, capture media, and record location data in low-connectivity environments. On the administrative side, ODK Central handles workspace security, user roles, form templates, and data exports to external analytical software. Although ODK excels at rugged field data collection, it omits several core capabilities vital for commercial out-of-home advertising research. Built-in functionality does not extend to billboard and campaign registries, geofenced attendance tracking, specialized market-research tools, mobile AI processing, image-quality enhancement, pitch analysis, automated dashboard generation, or coordinated research and coding workflows [3].

The main features of the Application are listed below:

- Android-based field data collection.
- Offline form completion and storage.
- Structured forms with logic and validation.
- Location and geospatial data collection.
- Image, audio, and video capture.
- Form and submission management.
- User roles and permissions within the proposed system.
- Centralized submission review and external analysis connections.

The limitations of the application are listed below:

- ODK provides general-purpose field collection rather than a dedicated billboard and campaign-management workflow.

- The reviewed documentation does not establish a built-in geofence-based attendance system linked to assigned campaign locations.
- The documented location functionality captures geographic information but does not establish the proposed hierarchical location and attendance workflow.
- The reviewed documentation does not establish the proposed market-research-specific questionnaire controls and campaign evidence structure.
- The reviewed documentation does not establish mobile-side AI processing for image-quality assessment or real-time billboard analysis.
- The reviewed documentation does not establish automated market-research dashboard generation and AI-generated insights as part of the standard collection workflow.
- The reviewed documentation does not establish pitch analysis or field-performance measurement.
- The reviewed documentation does not establish coordinated research agents or a Coding Agent integrated with the collection and analysis workflow.

Table 2 provides a side-by-side comparison of the three reviewed market research platforms and the proposed system. The proposed system is presented as a separate column to clearly distinguish its scope and functionality from the existing platforms.

Table 2: Comparison of Reviewed Applications

Features	**Applications**			
	Qualtrics [1]	SurveyMonkey [2]	ODK [3]	Proposed System
Web Questionnaires	✓	✓	✓	✓
Offline Mobile Collection	✓	✓	✓	✓
Android Field Collection	✓	✓	✓	✓
Media Capture	✓	✓	✓	✓
Location Capture	✓	✗	✓	✓
User Permissions	✓	✓	✓	✓
Analytics Dashboard	✓	✓	✗	✓
Billboard Registry	✗	✗	✗	✓
Multi-Market Management	✗	✗	✗	✓
Hierarchical Locations	✗	✗	✗	✓
Geofenced Attendance	✗	✗	✗	✓

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Table 2 – *Continued from previous page*

Features	Applications			
	Qualtrics [1]	SurveyMonkey [2]	ODK [3]	Proposed System
Visit Evidence	✗	✗	✗	✓
Offline-First SQLite	✗	✗	✗	✓
Incremental Synchronization	✗	✗	✗	✓
Media-Linked Records	✗	✗	✓	✓
Mobile AI Processing	✗	✗	✗	✓
Image Quality Analysis	✗	✗	✗	✓
Automated Insights	✗	✗	✗	✓
Field Performance Analysis	✗	✗	✗	✓
AI Agent Orchestration	✗	✗	✗	✓

10. Problem Statement

Market research teams face difficulties when they use separate tools for data collection, verification, and analysis. Traditional software supports simple survey forms, but field crews encounter problems when staff members record GPS coordinates, site photos, audio notes, and research findings in different applications. Separate storage systems force staff members to cross-reference multiple databases manually. Fragmented data systems also weaken the connection between raw field evidence and final market conclusions.

Managing data across disconnected tools creates heavy manual workloads for both field crews and office analysts. Researchers spend valuable hours matching site visits against specific billboard locations. Field staff frequently lose survey entries when mobile signals drop, leaving stray photos or audio files disconnected from their questionnaire records. Whenever information goes missing or fails to match, teams must stop and fix the errors manually before they can evaluate broader market trends. These recurring administrative bottlenecks slow down data verification, dashboard updates, and final client reporting.

Extended workloads and fragmented workflows will force organizations to rely heavily on manual coordination and repeated validation steps as research volume increases. The resulting heavy reliance will restrict the speed at which research can guide business decisions and will prevent teams from building consistent, reusable processes across different markets. Furthermore, this scattered approach will block companies from adopting auto-

mated artificial intelligence for field performance tracking, pitch evaluation, and real-time evidence processing. Consequently, valuable market insights will remain trapped behind slow, manual reviews.

11. Scope

The proposed system will cover the design, development, testing, and deployment of a web-based management and analytics platform with an Android application. It will support multiple market-research activities across markets and locations, including field collection, offline synchronization, geofenced attendance, media processing, AI-assisted analysis, competitor research, field-performance measurement, and automated reporting.

Module: User Management

- Company workspace creation and management.
- User registration and authentication.
- Role-based access and permissions.
- Device registration and authorization.
- Workflow-based information access.
- Account activity and audit records.

Module: Questionnaire Management

- Market-research questionnaire creation.
- Support for retail, billboard, competitor, and field surveys.
- Question branching and response validation.
- English and Urdu questionnaire support.
- OTP and eligibility controls.
- Questionnaire version management.

Module: Campaign Management

- Multi-market campaign creation and management.
- Billboard and advertisement registration.
- Competitor campaign and advertisement records.
- Hierarchical location management.
- Field assignment and campaign scheduling.
- Campaign, location, and evidence linking.

Module: Attendance Verification

- Field-agent location assignment.
- Geofence-based attendance verification.
- GPS-based visit validation.
- Visit timestamp recording.
- Location-based attendance records.
- Supporting visit evidence management.

Module: Data Synchronization

- SQLite-based offline data persistence.
- Incremental data synchronization.
- Submission queues and retry handling.
- Synchronization status tracking.
- Primary collection ID generation and management.
- Media association using collection IDs.

Module: Field Data-Collection

- Android-based field data collection.
- Assigned campaign and questionnaire access.
- Offline response and observation capture.
- GPS and timestamp collection.
- Photograph and audio capture.
- Secure local submission storage.

Module: AI Verification

- AI-based response quality checking.
- Evidence consistency verification.
- On-device preliminary AI processing.
- Real-time billboard and image analysis.
- Detection of missing or inconsistent information.
- Human review routing for uncertain results.

Module: Market Intelligence

- Secondary market-data collection.
- Competitor information collection.
- Web and document information extraction.

- Market and competitor comparison.
- Source and evidence tracking.
- Multi-market research analysis.

Module: Media Processing

- Image capture and management.
- Audio capture and management.
- Image quality assessment.
- Image enhancement where applicable.
- Audio quality and preprocessing.
- Media metadata and collection linking.

Module: Agent Orchestration

- Research planning agent coordination.
- Data collection and extraction agent coordination.
- Verification and analysis agent coordination.
- Automated reporting and dashboard preparation.
- Coding Agent support for development tasks.
- Agent permissions, workflow control, and human-review handling.

Module: Field Performance

- Field-agent activity tracking.
- Visit and task completion measurement.
- Evidence-completeness monitoring.
- Pitch recording and analysis.
- AI-assisted performance feedback.
- Performance scores and organization-defined rewards.

Module: AI-Agent Analytics

- Interactive market-data visualization.
- Campaign and location comparison.
- Market and competitor comparison.
- Automated AI-generated insights.
- Dashboard-based research reporting.
- Role-based analytics views.

12. Significance

The proposed system will make billboard research easier to trace from the initial question to the final result. Researchers will be able to inspect which questionnaire version produced a measure, which records were excluded, and which source supports a statement. Field staff will have a defined route for submitting observations, while analysts will work with campaign-linked evidence and consistent calculation rules. The system will also provide documented calculations, source references, review records, and controlled reporting for authorised users.

The proposed system will provide research teams with a coordinated workflow that reduces repeated movement of information between questionnaires, spreadsheets, media folders, evidence records, and reports. Companies will be able to manage campaign information, field observations, audience responses, and supporting sources within defined workflows. The system will also provide a basis for comparing manual, single-agent, and multi-agent approaches through measured task performance. Its organisational value will therefore be assessed through evidence rather than assumed time or cost savings.

The proposed system represents an evaluated implementation of specialised agents within a constrained market research workflow. It examines how coordinated agents support selected research tasks while preserving human review when evidence is disputed or operations fail. Numerical correctness, source support, abstention, and recovery from failures are evaluated separately so that generated explanations remain connected to their supporting evidence. Consent, clear questionnaire wording, and visible sampling limitations also support responsible research and clearer interpretation of findings.

13. Tools and Techniques

The proposed system uses web, mobile, backend, database, artificial intelligence, data processing, evidence retrieval, and security technologies. The selected tools support research planning, offline field collection, evidence management, data analysis, and report generation. The selected technologies also support the implementation and evaluation of the system's functional modules effectively.

- **Frontend Tools:** React and TypeScript will provide the company portal, questionnaire website, interactive dashboards, and reporting interfaces. Flutter and Dart will provide the Android field application, including offline questionnaires, local submission queues, and synchronisation functions.
- **Backend Tools:** The backend services will use Python for application logic and data processing. The backend services will use FastAPI for authenticated requests and operations. The backend services will handle campaigns, questionnaires, submissions, media, research runs, evidence, analysis, and reporting. The backend

services will use REST APIs to connect the web application and mobile application.

- **Database and Storage:** The database system will use PostgreSQL to store company, campaign, billboard, questionnaire, submission, claim, source, and report records. The mobile application will use SQLite to store queued field submissions on Android devices. The storage system will use object storage for permitted photographs and documents. The evidence retrieval system will use pgvector to retrieve authorised evidence.
- **AI Techniques:** LangGraph will coordinate bounded research agents through defined states for planning, collection, extraction, quality review, verification, analysis, and reporting. Retrieval-augmented generation will connect responses to authorised evidence, while controlled prompts, tool permissions, execution limits, and human review points will constrain agent operations.
- **Data Processing:** The data processing tools will include pandas and NumPy. These tools will support data cleaning, transformation, aggregation, and indicator calculations. The statistical analysis tools will include SciPy for selected statistical procedures. The analytical functions will calculate campaign measures and provide reference results for system evaluation.
- **Evidence Retrieval:** Approved public web sources and user-provided documents will be processed through structured extraction and retrieval techniques. Source identifiers, retrieval dates, passages, dataset versions, and claim-to-source links will be retained so generated findings can be traced to supporting evidence.
- **Security and Testing:** The security controls will include role-based access control, company-level permissions, authenticated requests, app-scoped device identifiers, audit records, and protected local storage. The development process will use Docker, Git, code review, reproducible test fixtures, and automated test cases for controlled development and system evaluation.

References

[1] Qualtrics, "Qualtrics Survey Platform," [Online]. Available: https://www.qualtrics.com/ [Accessed: Sep. 16, 2026].

[2] SurveyMonkey, "SurveyMonkey," [Online]. Available: https://www.surveymonkey.com/ [Accessed: Sep. 16, 2026].

[3] ODK, "ODK." [Online]. Available: https://getodk.org/ [Accessed: Sep. 16, 2026].